SWEET.TV Program Recommendation

LLM4Rec Midterm Quiz

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Temporal validation, reproducible experiments and an evidence-based recommendation strategy

Final strategy: hybrid_c0_a0.15. Development MAP@5: 0.41123. Frozen 12-week audit MAP@5: 0.39552.

Repository: https://github.com/Sasmas314/LLM4Rec-Midterm-2026

MISSING EVIDENCE: no verified private Kaggle score is available. Local metrics are not leaderboard scores.

1. Executive summary

The project predicts five program IDs for each of 1,260 users. It reconstructs preferences from 5,821,987 viewing records and the supplied broadcast schedule. A viewing event qualifies only when the union of watched intervals covers at least 80% of a broadcast. The resulting 611,758 events connect 4,841 active users to 3,865 programs.

Three qualifying recommendation families are implemented and evaluated individually: content-based filtering, association rules and implicit matrix factorization. Global popularity supplies an additional reference. Personal repeat frequency, recency, availability and hybrid variants test whether more complex signals improve the competition objective.

The selection criterion is mean MAP@5 across two development periods. It chooses hybrid_c0_a0.15 at 0.41123. Its 12-week audit score is 0.39552, versus 0.39734 for personal frequency. The difference is -0.00182, with a paired 95% bootstrap interval [-0.00634, +0.00257]. This does not demonstrate superiority on the longer horizon. The frozen selection remains the designated final model, and the report preserves the negative audit evidence.

Table 1. Representative methods, all evaluated on identical cohorts and scheduled candidates

Method	Development MAP@5	Audit MAP@5
Popularity	0.29339	0.32315
Personal frequency	0.40773	0.39734
Content-based	0.26570	0.26067
Association rules	0.29749	0.27878
Implicit ALS	0.28278	0.26352
Selected hybrid	0.41123	0.39552

SWEET.TV should start with a repeat-preference strategy, use a modest association signal for complementary suggestions, and keep personal frequency as a deployment challenger. A live A/B test would be required to establish engagement, retention or revenue impact.

2. Task, sources and data

The supplied presentation is the primary assignment specification. It requires at least two qualifying methods, a popularity baseline, temporal evaluation, analysis of the same two users, a GitHub link, a screenshot of the returned private score and an original unmodified AI-session export. It asks the student to submit the PDF and session export. Code and the exact final CSV belong in the repository.

The competition defines future preferences over weeks 31-42 and splits the public/private test by users, 50/50. The Data and Evaluation pages were inspected in the authenticated browser. Both the course and competition request five ranked program IDs. No hidden labels or server-side scoring source are supplied.

Table 2. Actual CSV files and measured row counts

Source file	Rows	Role
dataset11-30.csv	5,821,987	Viewing log
export_arh_11-20-final.csv	1,074,235	Broadcast schedule
export_arh_21-30-final.csv	1,074,394	Broadcast schedule
export_arh_31-42-final.csv	1,295,378	Broadcast schedule
submission.csv	1,260	Test template

Viewing logs contain 4,914 users and 202 channels. The complete schedule has 9,671 valid nonzero program IDs. Historical and target windows contain 236,772 and 144,713 fully contained annotated broadcasts respectively.

User IDs remain decimal strings when reading, joining and writing CSV files. Several exceed the signed 64-bit range and cannot safely pass through floating-point values. The loader maps vsetv_id to channel_id, stop_time to end_time and the source spelling duraton to duration. UTF-8 source titles and categories remain unchanged.

3. Preprocessing and interval matching

The log has 17 exact duplicate rows, 117 disagreements between declared duration and endpoint subtraction, and 0 invalid intervals. Endpoints govern watched time. The schedule audit records nulls, invalid durations and zero IDs per file; invalid IDs and intervals never become recommendation candidates.

All intervals use half-open boundaries [start,end). Both feeds have offset-free timestamps and share a source-local wall-clock assumption. The code does not invent a timezone or daylight-saving correction. Tests also cover equivalent timestamp-aware inputs.

For every user-channel pair, sort viewing intervals and union overlapping or adjacent intervals. This removes duplicated coverage across repeated or overlapping log rows. For each channel, sort broadcast starts and build a prefix maximum of broadcast end times. Two binary searches locate all broadcasts that can intersect each merged viewing interval. Every positive overlap contributes max(0, min(view_end, program_end) - max(view_start, program_start)) seconds.

Aggregate the overlap seconds per user-broadcast pair. Divide by full broadcast duration and retain ratios at least 0.8. Each qualifying broadcast contributes one frequency event. Multiple programs in a long session are handled separately. Partial watches accumulate across sessions. Gaps contribute no interaction. Conflicting broadcasts remain separately identified and are audited. Broadcasts crossing a train/validation cutoff are excluded from both segments, preventing coverage observed after the cutoff from entering training.

Mapping coverage before the 80% qualification rule is 50.47%: 2,746,573 of 5,441,773 merged viewing intervals overlap at least one valid nonzero annotated broadcast. The denominator includes channels without annotated schedule coverage. This is interval coverage, not the percentage of sessions that qualify as a complete program view.

Table 3. Interval reconstruction diagnostics

Matching audit	Measured value
merged_intervals_on_matched_channels	4,384,980
overlap_pieces	3,616,081
user_broadcast_overlaps	3,012,383
qualifying_events	611,758
qualifying_users	4,841
qualifying_programs	3,865
min_ratio	0.8
max_ratio	1.0

The matching cost is O(S log B + K) after sorting, for S merged intervals, B broadcasts per channel and K candidate overlaps examined. In pathological overlapping schedules K can be large. The implementation avoids a full session-broadcast Cartesian product.

The canonical event cache stores user_idx, broadcast_id, tv_show_id, watched_seconds, screen_time_ratio, start_time and end_time. A portable interaction table additionally exposes user_id, tv_show_id, interaction_weight and timestamp. Count weights match the competition target. Recency variants use 2^(-age_days / half_life), testing 7, 14, 28 and 56 days.

4. Exploratory analysis

There are 228,573 user-program pairs. Sparsity among active users and watched programs is 98.78%. The median active user has 82 qualifying events and 32 distinct programs. 73 logged users have no qualifying event, including 4 test users.

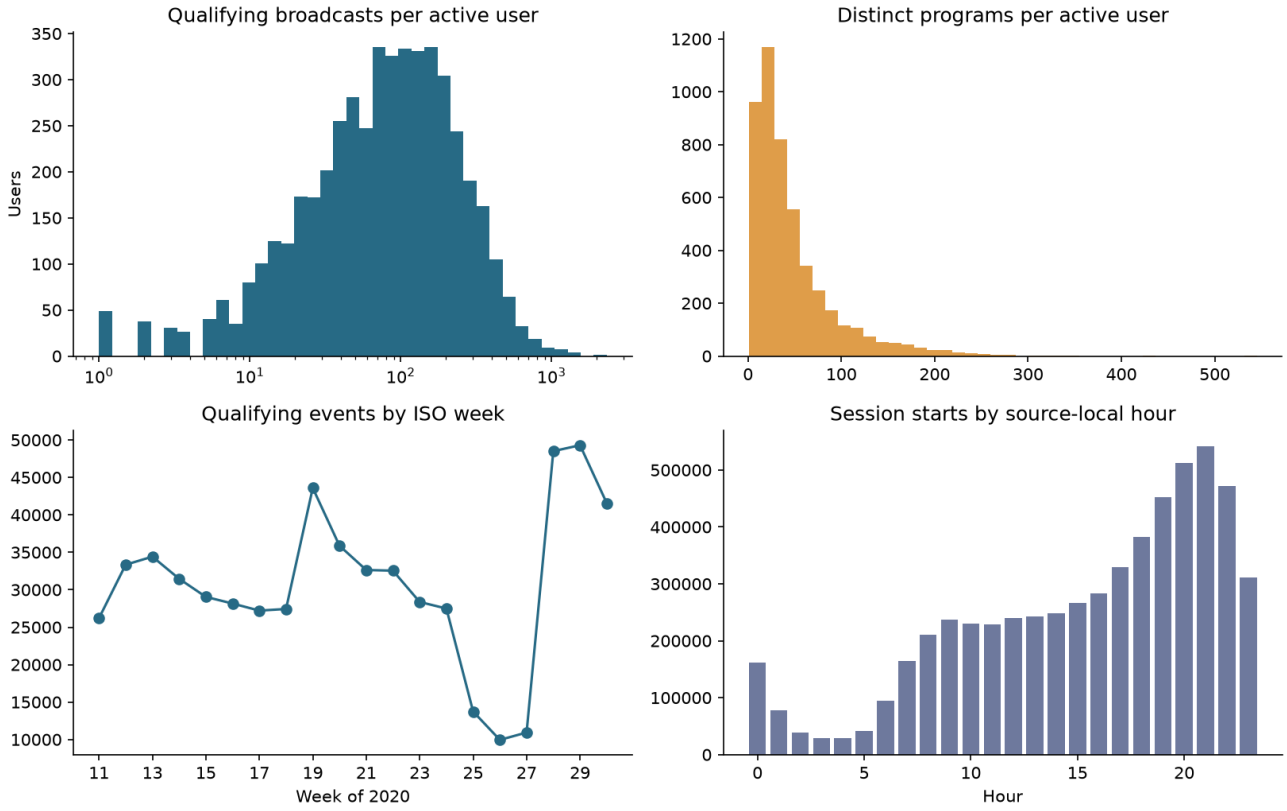


Figure 1. User activity, weekly qualifying events and source-local viewing times

The ten most frequent programs account for 41.79% of qualifying events. 852 programs have at most five viewers and contribute 2,372 events. Heavy concentration favors popularity while sparse tail evidence makes unregularized association rules unstable.

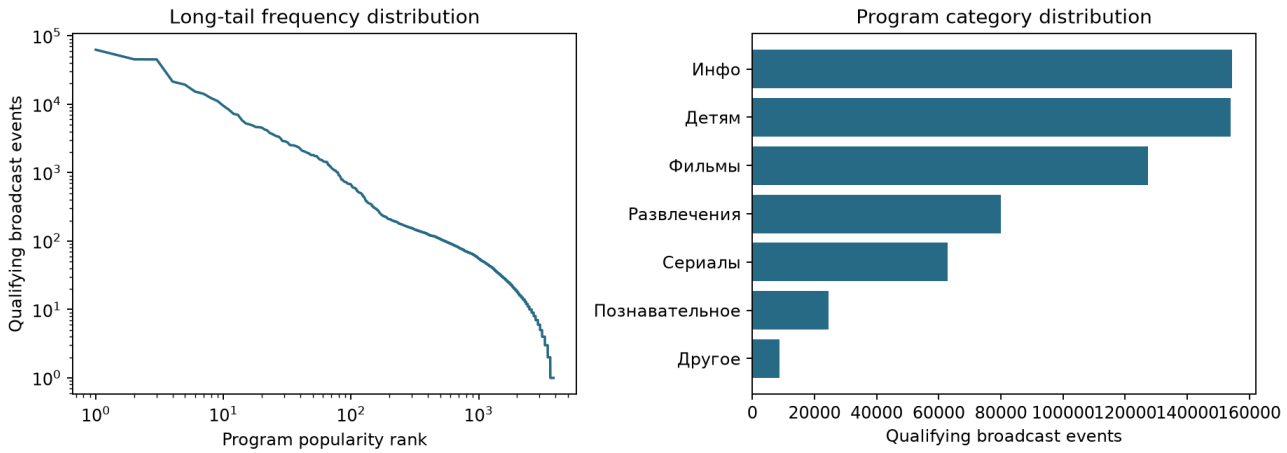


Figure 2. Program frequency and category distribution

5. Recommendation algorithms

Popularity and repeat frequency

Global popularity ranks candidate programs by training qualifying-event count. Personal frequency ranks each user's own counts and breaks ties by training popularity, then ascending program ID. Previously watched programs remain eligible because the target measures repeated future preferences. Cold users receive the same training-popularity fallback restricted to eligible scheduled programs.

Content-based filtering

TF-IDF uses title word unigrams/bigrams, category and genre tokens, plus actors, directors and production years when present. Metadata features receive weight 2 or 4, people/year features weight 0.5. Each item vector and the user's weighted aggregate profile receive L2 normalization. Cosine scores rank future candidates. Feature fitting uses the historical and target schedule, which is permitted catalog information. It never uses target user viewing behavior.

Association rules

Binary user-program incidence yields co-occurrence C_ij. Rule confidence is C_ij/support(i). The final edge is confidence × C_ij/(C_ij+5) / sqrt(support(j)/U), with support 2 or 5 and confidence 0.03 or 0.10. Diagonal edges are removed and only 120 outgoing neighbors are retained. User log(1+decayed qualifying frequency) weights the outgoing association scores. This supports discovery but can pull narrow profiles toward globally common audience habits.

Implicit matrix factorization

Alternating least squares minimizes a confidence-weighted implicit objective, with preference 1 for observed pairs and 0 for unobserved pairs. The implementation passes 10 log(1+decayed counts) for observed confidence. It tests 32 factors/regularization 0.1 and 64 factors/regularization 0.5, with 25 iterations, seed 42 and two worker threads. Counts are implicit positive signals, not explicit ratings. Unseen items lack learned preference evidence.

Hybrid and availability

The frequency component may multiply scores by ((future_airings+2)/(historical_airings+2))^p, where p is 0.25, 0.5 or 1. Each nonnegative component is divided by its row maximum before blending. Tests also include rank fusion and a channel/time signal. The selected model combines decayed, availability-adjusted repeat frequency with a small association term. Complete settings are recorded in experiment_specs.json and configs/best_model.yaml.

6. Temporal validation and metrics

Table 4. All training begins 9 March 2020, all end boundaries are exclusive

Split	Train end	Validation end	Days	Users with ≥5 labels
dev_4w	2020-04-06	2020-05-04	28	1039
dev_2w	2020-04-20	2020-05-04	14	799
audit_12w	2020-05-04	2020-07-27	84	1199

Two development periods share the end date 4 May and therefore overlap. Their unweighted mean is a tuning criterion, not two independent replications. The chosen configuration is frozen before the 4 May-27 July twelve-week audit. The audit matches the target horizon, though final training uses a longer twenty-week history and a different season. The previous local solution already contained these periods and results, so this rerun is a reproducibility audit, not a newly blinded study.

Every qualifying model uses the same candidate pool: nonzero IDs in broadcasts fully contained within the target period. No target ground-truth item filtering narrows the pool. Metadata from the target schedule is assumed known at recommendation time. An unfiltered-frequency diagnostic intentionally ignores availability and is excluded from the main equal-candidate comparison.

Table 5. Excluded and secondary cohorts remain explicit

Split	Active with 1-4 labels	No qualifying labels	Candidate label coverage
dev_4w	163	58	100.000%
dev_2w	346	115	100.000%
audit_12w	41	20	100.000%

The primary cohort contains template users with at least five distinct qualifying target programs. Ground truth is their five most frequent IDs, with numeric-ID tie breaking. Active users with fewer labels are also evaluated using their actual labels. Zero-label users have undefined AP and are counted separately. Such cohort conditioning can make the primary score optimistic for all-user deployment.

AP@5 = Σ[r=1..5] Precision@r × relevant_hit_at_r / min(5, number_of_relevant_items). Duplicate predictions cannot create extra hits. MAP@5 averages AP over the declared cohort. Precision divides hits by five. Recall divides by the true relevant-set size. HitRate is 1 if any recommendation hits. With exactly five labels, precision and recall coincide. Binary MAP distinguishes hit positions, but not permutations of five correct items. Ordered agreement and graded NDCG are separate diagnostics because the competition text also emphasizes order. The hidden scoring implementation remains unavailable.

7. Measured results and model selection

Table 6. Frozen twelve-week audit results

Method	MAP@5	P@5	R@5	HitRate@5
Popularity	0.32315	0.40400	0.40400	0.92077
Personal frequency	0.39734	0.46088	0.46088	0.93244
Content-based	0.26067	0.29258	0.29258	0.81735
Association rules	0.27878	0.37465	0.37465	0.90409
Implicit ALS	0.26352	0.36981	0.36981	0.86822
Selected hybrid	0.39552	0.45805	0.45805	0.92327

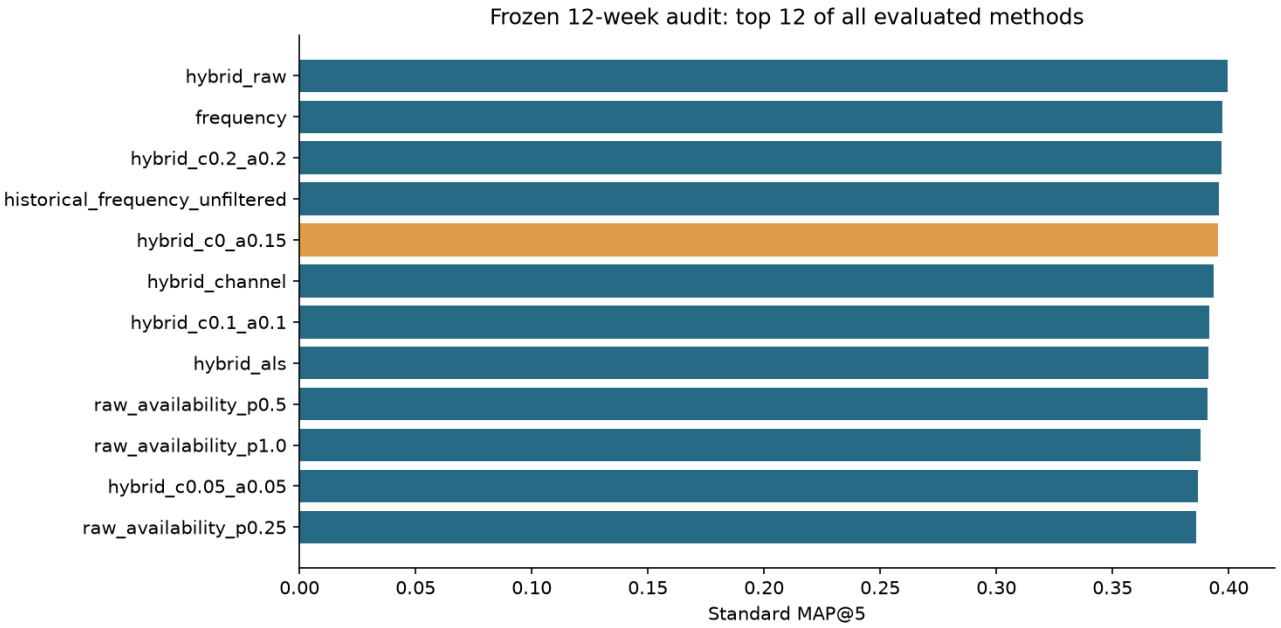


Figure 3. Audit ranking, including diagnostic variants. Highlight identifies the frozen development winner

The selected model scores 0.41123 on development. Personal frequency scores 0.40773, a difference of +0.00350. On the audit, the selected strategy scores 0.39552; personal frequency scores 0.39734. The audit's numerical winner is hybrid_raw at 0.39967. The selected configuration is not replaced using this audit outcome.

The paired audit delta against frequency is -0.00182; the 95% interval [-0.00634, +0.00257] uses 2,000 user bootstrap resamples, seed 42. It reflects this cohort and period, not season changes, repeated model searches or uncertainty in hidden labels.

8. Optimization, cost and scalability

31 configurations are evaluated on all three periods. The search covers recency, restricted history, availability exponents, content weighting, association support/confidence, ALS factors/regularization and hybrid weights. Each standalone family is evaluated before hybrid selection. Full parameters and metrics are in results/experiments.csv. The strongest standalone qualifying family is identified by development mean rather than audit rank.

The strongest qualifying configuration on development is association_s5_c0.1 (0.29791). Association variants test meaningful support/confidence changes. No unbounded parameter search or Kaggle-driven search is performed.

Table 7. Measured audit-stage wall time and resident process memory

Method	Fit + scoring (s)	Cached rank (s)	Process RSS (MiB)
Popularity	0.007	0.060	794
Personal frequency	0.010	0.071	794
Content-based	1.070	0.686	849
Association rules	0.096	0.116	909
Implicit ALS	0.474	0.186	1011
Selected hybrid	0.063	0.153	976

Times use a shared engine: fit + scoring includes a component build only when it is absent from the cache, and cached rank includes score copying, ranking and minor score operations. They are incremental workflow costs, not isolated cold-start benchmarks. RSS is process memory sampled after each method, not model-only or peak memory. ALS uses two workers, BLAS one thread and Polars four. Hardware: 20 logical CPUs, approximately 31.1 GiB RAM, AMD Radeon graphics; no GPU training is used.

Frequency training is linear in event count. Sparse association construction costs approximately the sum of squared user profile sizes before neighbor truncation; broad users can densify co-occurrence. The dense similarity intermediate therefore needs partitioning at larger catalog sizes. Content scoring can use sparse matrix multiplication or an approximate nearest-neighbor index. ALS stores O((U+I)f) factors and supports batched dot-product retrieval. The current implementation caches user-by-candidate score matrices and sorts the full candidate list. Production inference should batch users, retain sparse edges, select Top-K without full sorts and monitor memory.

9. Two-user study: historical preferences

The same two users appear in every comparison. Selection uses permissible full historical behavior, without target outcomes. Case A maximizes Top-5 concentration among test users with at least 200 qualifying broadcasts and eight programs. Case B minimizes it among users with at least 100 events and thirty programs. These deliberately contrasting cases are explanatory, not a representative sample.

User A: 17749612088451113386

569 events, 37 distinct programs, Top-5 concentration 93.7%, top-program share 57.6%. Leading channel: Nickelodeon HD. Leading first-genre: комедия.

Table 8. User A, observed historical Top-5

Program ID	Original program title	Category	Views
12001682	М/с 'Губка Боб Квадратные Штаны'.	Детям	328
12002856	М/с 'Элвиннн!!! и бурундуки'.	Детям	132
12001734	М/с 'Волшебные покровители'.	Детям	57
12001732	М/с 'Пингвины из 'Мадагаскара'.	Детям	11
10001336	Т/с 'Кости', 5 сезон, 13 с. 'Дантист в траншее'.	Сериалы	5

Category mix by qualifying events: Детям 93.5%, Фильмы 3.5%, Сериалы 1.8%.

User B: 4328761455849723043

310 events, 239 distinct programs, Top-5 concentration 9.0%, top-program share 2.9%. Leading channel: ViP Comedy CEE. Leading first-genre: боевик.

Table 9. User B, observed historical Top-5

Program ID	Original program title	Category	Views
240081	Факты.	Инфо	9
200337	Утро с Интером.	Развлечения	6
10001400	Т/с 'Доктор Кто', 12 сезон, 3 с. 'Сирота-55'.	Сериалы	5
2400508	Сегодня.	Инфо	4
10001261	Т/с 'Мыслить как преступник'.	Сериалы	4

Category mix by qualifying events: Фильмы 69.7%, Развлечения 7.4%, Сериалы 6.8%.

10A. User A: model Top-5

Each row is a genuine future prediction. Titles and categories come from the supplied schedule. The interpretation indicates the model mechanism, not proof of user satisfaction.

Popularity

#	Program ID	Title	Category	Past views
1	2400480	ТСН.	Инфо	1
2	12001682	М/с 'Губка Боб Квадратные Штаны'.	Детям	328
3	240081	Факты.	Инфо	1
4	20088	'Завтрак с 1+1'. Информационно-развлекательная программа.	Развлечения	0
5	2400508	Сегодня.	Инфо	0

Training frequency across all users drives this list. 3/5 recommendations have qualifying historical views.

Personal frequency

#	Program ID	Title	Category	Past views
1	12001682	М/с 'Губка Боб Квадратные Штаны'.	Детям	328
2	12002856	М/с 'Элвиннн!!! и бурундуки'.	Детям	132
3	12001734	М/с 'Волшебные покровители'.	Детям	57
4	12001732	М/с 'Пингвины из 'Мадагаскара'.	Детям	11
5	10001336	Т/с 'Кости', 5 сезон, 13 с. 'Дантист в траншее'.	Сериалы	5

The user's own qualifying-event counts drive this list. 5/5 recommendations have qualifying historical views.

Content-based

#	Program ID	Title	Category	Past views
1	12001682	М/с 'Губка Боб Квадратные Штаны'.	Детям	328
2	120078041	М/ф 'Железяки'.	Детям	0
3	12001734	М/с 'Волшебные покровители'.	Детям	57
4	12002378	М/с 'Удивительный мир Гамбола'.	Детям	0
5	12001820	М/с 'Котопёс'.	Детям	0

Cosine similarity to the user's metadata profile drives this list. 2/5 recommendations have qualifying historical views.

Association rules

#	Program ID	Title	Category	Past views
1	2400480	ТСН.	Инфо	1
2	240081	Факты.	Инфо	1
3	2400467	'Викна-Новости'.	Инфо	1
4	2400508	Сегодня.	Инфо	0
5	12001682	М/с 'Губка Боб Квадратные Штаны'.	Детям	328

Programs shared by overlapping audiences receive rule support. 4/5 recommendations have qualifying historical views.

Implicit ALS

#	Program ID	Title	Category	Past views
1	12001682	М/с 'Губка Боб Квадратные Штаны'.	Детям	328
2	12002856	М/с 'Элвиннн!!! и бурундуки'.	Детям	132
3	12001734	М/с 'Волшебные покровители'.	Детям	57
4	400468	Аферисты в сетях, 5 сезон, 1 эп.	Познавательное	2
5	240081	Факты.	Инфо	1

Latent factors represent patterns of implicit viewing. 5/5 recommendations have qualifying historical views.

Selected hybrid

#	Program ID	Title	Category	Past views
1	12001682	М/с 'Губка Боб Квадратные Штаны'.	Детям	328
2	12002856	М/с 'Элвиннн!!! и бурундуки'.	Детям	132
3	12001734	М/с 'Волшебные покровители'.	Детям	57
4	2400480	ТСН.	Инфо	1
5	240081	Факты.	Инфо	1

Repeat frequency, scheduled opportunities and association evidence combine. 5/5 recommendations have qualifying historical views.

10B. User B: model Top-5

Each row is a genuine future prediction. Titles and categories come from the supplied schedule. The interpretation indicates the model mechanism, not proof of user satisfaction.

Popularity

#	Program ID	Title	Category	Past views
1	2400480	ТСН.	Инфо	2
2	12001682	М/с 'Губка Боб Квадратные Штаны'.	Детям	1
3	240081	Факты.	Инфо	9
4	20088	'Завтрак с 1+1'. Информационно-развлекательная программа.	Развлечения	2
5	2400508	Сегодня.	Инфо	4

Training frequency across all users drives this list. 5/5 recommendations have qualifying historical views.

Personal frequency

#	Program ID	Title	Category	Past views
1	240081	Факты.	Инфо	9
2	200337	Утро с Интером.	Развлечения	6
3	10001400	Т/с 'Доктор Кто', 12 сезон, 3 с. 'Сирота-55'.	Сериалы	5
4	2400508	Сегодня.	Инфо	4
5	10001261	Т/с 'Мыслить как преступник'.	Сериалы	4

The user's own qualifying-event counts drive this list. 5/5 recommendations have qualifying historical views.

Content-based

#	Program ID	Title	Category	Past views
1	90014009	Х/ф 'Спина к спине'.	Фильмы	0
2	90049583	Х/ф 'Шафт'.	Фильмы	0
3	90050940	Х/ф 'Сават'.	Фильмы	0
4	90076845	Х/ф 'Спираль'.	Фильмы	0
5	90074210	Х/ф 'Фрираннер'.	Фильмы	0

Cosine similarity to the user's metadata profile drives this list. 0/5 recommendations have qualifying historical views.

Association rules

#	Program ID	Title	Category	Past views
1	2400480	ТСН.	Инфо	2
2	240081	Факты.	Инфо	9
3	200469	Варьяты. Лучшее.	Развлечения	2
4	6700482	Секретные материалы.	Другое	0
5	5900347	'Деньги'.	Инфо	1

Programs shared by overlapping audiences receive rule support. 4/5 recommendations have qualifying historical views.

Implicit ALS

#	Program ID	Title	Category	Past views
1	1100411	'Вещдок'.	Познавательное	1
2	10002454	Т/с 'Гримм', 1 сезон, 15 с.	Сериалы	0
3	2500415	'Месть природы'.	Инфо	1
4	90074576	Х/ф 'Призрачный гонщик 2'.	Фильмы	1
5	90081719	Х/ф 'Патрик'.	Фильмы	2

Latent factors represent patterns of implicit viewing. 4/5 recommendations have qualifying historical views.

Selected hybrid

#	Program ID	Title	Category	Past views
1	240081	Факты.	Инфо	9
2	10001261	Т/с 'Мыслить как преступник'.	Сериалы	4
3	2400480	ТСН.	Инфо	2
4	20088	'Завтрак с 1+1'. Информационно-развлекательная программа.	Развлечения	2
5	90070747	Х/ф 'Затерянный мир'.	Фильмы	3

Repeat frequency, scheduled opportunities and association evidence combine. 5/5 recommendations have qualifying historical views.

11. User experience and observed failures

Case A is strongly concentrated in children's programming. SpongeBob SquarePants (12001682) is the most repeated historical ID. The association model can still promote TSN (2400480) and Fakty (240081), showing how shared audiences can inject news into a narrow children's profile. Household sharing is a plausible mechanism, not a measured demographic fact. The hybrid retains the repeat signal but can preserve some of that contamination.

Case B has a much broader catalog and low repeat concentration. Frequently repeated news IDs can dominate a frequency objective even when the broader channel/genre history suggests varied films. Content similarity supplies plausible discovery candidates, while their low per-ID repeat evidence makes them uncertain predictions of the five most frequent future programs. Completed films may not be watched again. These are hypotheses about recommendation utility; no future test-user outcomes are visible.

Table 22. Observed cold and sparse-profile failure analysis

Audit history cohort	Users	Method	MAP@5
cold_0	13	Hybrid	0.21436
cold_0	13	Frequency	0.21436
sparse_1_to_5	65	Hybrid	0.15672
sparse_1_to_5	65	Frequency	0.17062
medium_6_to_20	566	Hybrid	0.35431
medium_6_to_20	566	Frequency	0.36080
broad_over_20	555	Hybrid	0.46975
broad_over_20	555	Frequency	0.46544

There are 3,783 future scheduled IDs without qualifying historical interactions, including 1,435 IDs absent from the historical schedule. Pure collaborative factors cannot learn these programs. Content can describe them, but the offline metric supplies weak incentives for discovery. Candidate coverage, recommendation novelty and catalog concentration should accompany accuracy in deployment.

12. Business decision and limitations

For this submission, use the frozen development-selected hybrid. Its frequency backbone fits the explicitly defined repeat-viewing target. The modest association term allows complementary audience evidence, while availability removes impossible programs. The long-horizon audit does not establish an advantage over personal frequency, so a production decision should retain the simpler baseline as a serious challenger and evaluate the added component prospectively.

Global popularity is inexpensive and useful for empty profiles, but lacks individual preferences. Content makes metadata explanations and cold-item suggestions possible, yet broad category similarity does not reliably predict repeated per-ID preferences. Association rules are interpretable but inherit audience concentration and co-occurrence sparsity. ALS compresses the interaction matrix, but the tested configurations underperform frequency and require additional training and operational complexity.

A practical deployment would update histories incrementally, check schedule availability before serving, batch score retrieval, cache repeated requests and log fallback rates. Monitor accuracy by activity cohort, exposure concentration and long-tail coverage. Reserve a separate experimentally controlled discovery surface if the business objective extends beyond forecasting habitual viewing. No revenue, retention or satisfaction improvement is demonstrated by these offline results.

No hidden test labels or server-side metric implementation are available. The assignment AP convention is implemented and hand-tested; target tie-breaking and organizer matching details remain assumptions.

Overlapping development windows are correlated. The previous project exposed the audit results before this rerun, so it cannot serve as a newly blinded holdout. No retuning uses the rerun audit.

Source clocks have no timezone offsets. Four annotated overlaps and extensive missing schedule IDs can distort coverage. Counting a broadcast with at least 80% coverage loses partial-interest information.

Temporal cohorts exclude users with no qualifying labels and primarily require five targets. Small cold cohorts cannot support broad claims. Schedule availability is assumed known, while actual user availability and exposure are unobserved.

Program IDs can aggregate episodes. Metadata is incomplete, and the modal source title is descriptive rather than a guaranteed future episode title. The model learns accounts/devices, which may represent households.

Search is limited, especially for ALS. Timings use shared caches, memory is sampled rather than profiled at peak, and the implementation has not been load-tested at production scale.

13. Reproducibility and deliverables

https://github.com/Sasmas314/LLM4Rec-Midterm-2026

Use Python 3.12. Install requirements.txt, place the five original CSVs together, then run: python -m pytest tests -q; python run_pipeline.py --stage all --data-dir PATH --force; python scripts/finalize_artifacts.py --data-dir PATH; python scripts/generate_report.py. Commands are separate invocations in the README. Input SHA-256 hashes bind the processed caches to their sources. Raw data is excluded from Git.

Code reuses and extends a prior local solution found in Documents/ChatGPT/HSE/sweet_tv_exam. This run independently rebuilds raw preprocessing, interval matches and all experiment predictions. It adds HitRate, cost measurements, additional case-study methods, a PDF, export preservation and external submission evidence. It does not claim authorship of an entirely new pipeline or a newly blinded study.

The root submission.csv is an exact copy of outputs/submission.csv. The automated validator checks schema, row order, full user coverage, five positive unique IDs and membership in the final schedule. results/experiments.csv records every configuration. final_model_metrics.json distinguishes local validation from returned Kaggle scores.

The session-export directory preserves a byte-for-byte copy of the actual Codex JSONL record, with SHA-256 manifest. Because this chat is still running when it copies the file, the snapshot may omit the last tool calls and final response. Refresh after the chat finishes using scripts/refresh_session_export.ps1. The file is not translated, sanitized or reconstructed. The instructor has not confirmed whether native Codex JSONL satisfies any OpenCode-specific export requirement. The unrelated WARNING_BANNER session.json is not presented as evidence.

14. Kaggle submission evidence

MISSING EVIDENCE: no verified returned private score or score screenshot. The prepared CSV is valid locally; this does not prove Kaggle accepted or scored it. Validated CSV ready. Browser upload blocked until extension file URL access is enabled.

Kaggle's page allows up to two final selections and may auto-select public-score candidates if fewer are selected. The course grades the returned private score of the designated submission rather than frozen leaderboard position. A genuine screenshot and matching CSV checksum are needed to connect the claimed score to the exact artifact.

15. References and experiment appendix

Jin, S. LLM4Rec. Midterm Quiz (1).pptx. Supplied course assignment, slides 2-9.

ogurbych (2020). sweet.tv - TV program recommender. Kaggle.
https://www.kaggle.com/competitions/sweettv-tv-program-recommender

Kaggle competition Data and Evaluation pages, inspected 24 October 2026.
https://www.kaggle.com/competitions/sweettv-tv-program-recommender/data

Implicit ALS implementation: implicit.cpu.als.AlternatingLeastSquares. Dependency version recorded in requirements.txt.
Source: https://github.com/benfred/implicit

Codex export documentation: https://learn.chatgpt.com/docs/developer-commands. The /export command creates a Markdown export after a turn completes. Native JSONL is preserved separately in this submission.

16. All measured MAP@5 results

Table 23. Complete experiment set. Unfiltered frequency is a separate candidate-constraint diagnostic

Configuration	Dev 4w	Dev 2w	Audit 12w
als_32_r0.1	0.31792	0.24763	0.26352
als_64_r0.5	0.30002	0.22811	0.22676
association_s2_c0.03	0.31348	0.28150	0.27878
association_s5_c0.03	0.31215	0.28118	0.27783
association_s5_c0.1	0.31384	0.28197	0.27871
availability_p0.25	0.40496	0.38846	0.37955
availability_p0.5	0.40569	0.38671	0.37987
availability_p1.0	0.40355	0.37030	0.37190
channel_time	0.38579	0.36791	0.37912
content	0.27311	0.25829	0.26067
content_metadata4	0.25698	0.24325	0.24781
frequency	0.42024	0.39521	0.39734
global_popularity	0.30732	0.27945	0.32315
historical_frequency_unfiltered	0.41241	0.39023	0.39574
hybrid_als	0.41529	0.39722	0.39115
hybrid_c0.05_a0.05	0.41358	0.39409	0.38673
hybrid_c0.15_a0	0.40169	0.38566	0.37905
hybrid_c0.1_a0.1	0.41837	0.39806	0.39162
hybrid_c0.2_a0.2	0.42206	0.39929	0.39691
hybrid_c0_a0.15	0.42201	0.40045	0.39552
hybrid_channel	0.41766	0.39694	0.39341
hybrid_raw	0.41947	0.39634	0.39967
last28_frequency	0.42024	0.38967	0.37146
rank_fusion	0.40510	0.39192	0.37694
raw_availability_p0.25	0.40429	0.38560	0.38587
raw_availability_p0.5	0.40499	0.38482	0.39103
raw_availability_p1.0	0.40347	0.37412	0.38791
recency_14d	0.39624	0.37990	0.35217
recency_28d	0.39947	0.38731	0.37311
recency_56d	0.39807	0.38597	0.37966
recency_7d	0.37446	0.35174	0.30567